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% Clean up the Command Window
clear;
close all;
clc;

T3 = 400;
n = zeros(1,length(T3));

for z = 1:length(T3)

% Set up Boundary Conditions

% Temperature Bounds:

T_Cond_in = 11.6;           % Coolant Inlet Temperature (Celcius)
T_Cond_out = T_Cond_in + 3; % Coolant Outlet Temperature (Celcius)

% Givens:

Q_Boiler = 30E3;           % Thermal Energy in Through Boiler (MW)

% State 1
T1 = T_Cond_out + 20;      % Saturation Temperature (Celcius)
s1 = XSteam('sL_T',T1);   % Specific Entropy
v1 = XSteam('vL_T',T1);   % Specific Volume
h1 = XSteam('hL_T',T1);   % Specific Enthalpy
p1 = XSteam('psat_T',T1);

% State 2
s2 = s1;                  % Specific Entropy (Isentropic)
p2 = 10;                  % Pressure(Bars) @ State 2: Heat Exchanger
T2 = XSteam('T_ps',p2,s2); % State 2 Temp (Celcius)
v2 = v1;
h2 = XSteam('h_ps',p2,s2);

% State 3
p3 = p2;                  % Pressure(Bars) @ State 3: Turbine
h3 = XSteam('h_pT',p3,T3(z)); % Specific Enthalpy
s3 = XSteam('s_pT',p3,T3(z)); % Specific Entropy

% State 4
s4s = s3;                % Specific Entropy (Isentropic)
T4 = XSteam('Tsat_p',p1); % Boundary Condition
sf = XSteam('sL_T',T4);
sfg = XSteam('sV_T',T4) - sf;
x = (s4s-sf)/sfg;        % Calculate Quality to get isentropic enthalpy
hf = XSteam('hL_T',T4);
hfg = XSteam('hV_T',T4) - hf;
h4s = hf + x*hfg;        % Calculate Isentropic Enthalpy

%Turbine

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isoTurbineEff = 0.9;
wTs = h3 - h4s;
wT = isoTurbineEff*wTs;
h4 = h3 - wT;

%Boiler
q_Boiler = h3-h2;
m = Q_Boiler/q_Boiler;

%Condenser
q_Condenser = h1-h4;

%Pump
wPump = h2 - h1;

% Coolant Loop

% Number of Tubes

L = 10;
D = 25E-3;
h = 1000;
rho = 999;
v = 1.12E-6;
eps = 45E-6;
delT1 = T4 - T_Cond_in;
delT2 = T4 - T_Cond_out;

Tlim = (delT1-delT2)/log(delT1/delT2);

N = (-m*q_Condenser*10^3)/(h*pi*L*D*Tlim);

mCoolant = m/N;

vCoolant = mCoolant/(pi*0.25*D*rho);

Re = vCoolant*D/v;

%f = (-1.8*log10(((eps/D)/3.7))^1.11 + 6.9/Re)^-1;
f = 64/Re;

FrictLoss = (f*rho*L*vCoolant^2)/(2*D);

wPump2 = m*FrictLoss;

% Overall Efficiency

sumWork = m*(wT - wPump) - wPump2;

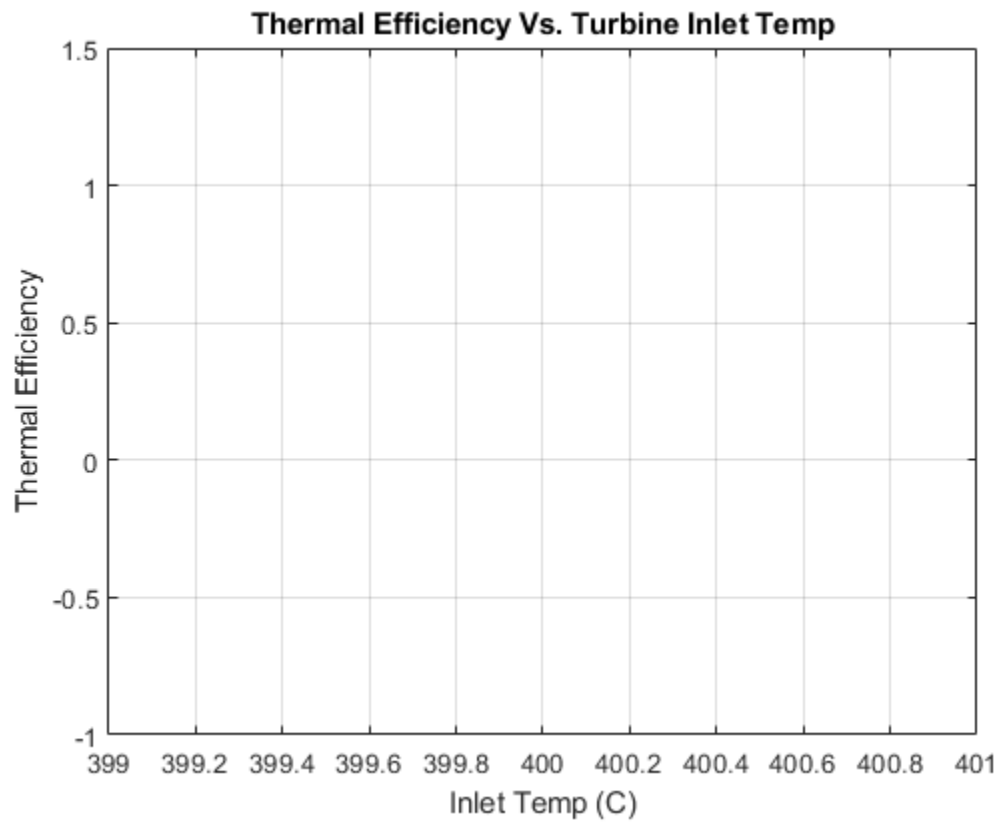
n(z) = sumWork/Q_Boiler;

end

plot(T3, n);

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title('Thermal Efficiency Vs. Turbine Inlet Temp');  
xlabel('Inlet Temp (C)');  
ylabel('Thermal Efficiency');  
grid on
```



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